

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$F: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right) \quad \text{gradient}$$

$$\nabla \cdot F(x) = \frac{\partial F_1}{\partial x_1}(x) + \dots + \frac{\partial F_n}{\partial x_n}(x) \quad \text{divergence}$$

$$\nabla \times F(x) = \underset{\wedge_{n=3}}{\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{F_2}{\partial x} - \frac{F_1}{\partial y} \right)$$

$$\Delta f = \frac{\partial^2 f}{\partial x_1^2} + \dots + \frac{\partial^2 f}{\partial x_n^2}$$

$$n=3$$

$$\nabla \times \nabla f = 0$$

$$\nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}, \dots, \dots \right) = \vec{0}$$

$$\nabla \cdot (\nabla \times F) = 0$$

$$\nabla \cdot (-y_z + z_y, z_x + x_z, -x_y + y_x) =$$

$$\cancel{z_y}_x - \cancel{y_z}_x + \cancel{x_z}_y - \cancel{z_x}_y + \cancel{y_x}_z - \cancel{x_y}_z = 0$$

cohomological complex, koje imaš dva

zaporeda: pozicije 0

$$\mathbb{R} \xrightarrow{d} \mathbb{R}^3 \xrightarrow{d} \mathbb{R}^3 \xrightarrow{d} \mathbb{R}$$

$$K = [0, 1] \times [0, 1] \subseteq \mathbb{R}^2$$

$$\int_K \nabla \cdot \vec{F} \, dx \, dy = \int_{\partial K} \vec{F} \, d\vec{n} \quad \text{normala}$$

$$\begin{aligned} \int_0^1 \int_0^1 \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dx \, dy &= \int_0^1 dy \int_0^1 \frac{\partial F_1(x, y)}{\partial x} dx + \int_0^1 dx \int_0^1 \frac{\partial F_2(x, y)}{\partial y} dy \\ &= \int_0^1 F_1(x, y) \Big|_0^1 dy + \int_0^1 F_2(x, y) \Big|_0^1 dx = \end{aligned}$$

$$= \int_0^1 (F_1(1, y) - F_1(0, y)) dy + \int_0^1 (F_2(x, 1) - F_2(x, 0)) dx$$

$$\int_{\partial K} \vec{F} \, d\vec{n} = \int_0^1 (F_1(x, 0), F_2(x, 0)) \begin{pmatrix} dx \\ 0 \end{pmatrix} + \dots$$

$$= \int_0^1 -F_2(x, 0) dx + \dots = \int_0^1 (F_1(1, y) - F_1(0, y)) dy + \int_0^1 (F_2(x, 1) - F_2(x, 0)) dx$$

$$u_{xy} = x^2 y \quad u(x, 0) = x \quad u(0, y) = y^2$$

$$u_x = \frac{1}{2} x^2 y^2 + c(x)$$

$$u = \frac{1}{6} x^3 y^2 + c(x) + D(y)$$

$$u(x, 0) = c(x) + D(0) = x$$

$$u(0, y) = c(0) + D(y) = y^2$$

$$c(x) + D(y) = x + y^2$$

$$c(0) + D(0) \text{ in both cases is } 0$$

$$u(x, y) = \frac{1}{6} x^3 y^2 + x + y^2$$

$$U_{xy} + U_x = y e^{-y}$$

$$u(0, y) = 0$$

$$u(x, x) = 0$$

↓

integrando p.o. x

$$u_y(0, y) = 0$$

$$\frac{u(x, x)}{\partial y} = 0$$

$$U_y + U = x y e^{-y} + C(y)$$

$$v(0, y) = 0$$

$$0 + 0 = 0 + C(y) = 0$$

$$U_y + U = x y e^{-y}$$

lineal en diferenciales exacta

homogen del

$$u + u_y = 0$$

$$u = -\frac{\partial u}{\partial y} \quad dy = -\frac{du}{u} \Rightarrow y = -\ln u + C(x)$$

$$u = C e^{-y}$$

particular

$$C'(y) e^{-y} = x y e^{-y}$$

$$C'(y) = x y$$

$$C(x, y) = \frac{1}{2} x y^2$$

$$u = \frac{1}{2} x y^2 e^{-y} + C(x) e^{-y}$$

$$u(0, y) = C(0) e^{-y} = 0$$

$$u(x, x) = \frac{1}{2} x^3 e^{-x} + C(x) e^{-x} = 0$$

$$\Rightarrow C(x) = -\frac{1}{2} x^3$$

$$u(x, y) = \frac{1}{2} x e^{-x} (y^2 - x^2)$$

c)

$$u_x + u_y = 0$$

$$u(x, 0) = x^2$$

$$u(x, y) = \sum_{i,j} a_{ij} x^i y^j$$

$$u(x, 0) = \sum_i a_{i,0} x^i = x^2 \Rightarrow a_{2,0} = 1$$

$$a_{i,0} = 0 \text{ za } i \neq 2$$

$$u_x = \sum_{i,j} a_{ij} i x^{i-1} y^j = a_{1,0} + \sum_{i,j \geq 0} a_{i+1,j} (i+1) x^i y^j$$

$$u_y = \sum_{i,j} a_{ij} j x^i y^{j-1} = a_{0,1} + \sum_{i,j \geq 0} a_{i,j+1} (j+1) x^i y^j$$

$$u_x + u_y = a_{0,1} + a_{1,0} + \sum_{i,j \geq 1} (a_{i+1,j} (i+1) + a_{i,j+1} (j+1)) x^i y^j = 0$$

$$\Rightarrow (a_{i+1})(i+1) + a_{i,j+1}(j+1) = 0 \text{ za } \forall x, y$$

$$a_{0,1} + a_{1,0} = 0 \Rightarrow a_{0,1} = -a_{1,0}$$

$$a_{2,1} = a_{1,2}$$

$$\Rightarrow a_{0,j} = 0 \text{ razem } a_{0,2} = -2$$

$$a_{1,1} = -a_{0,2}$$

$$a_{1,1} = -a_{2,0}$$

$$a_{1,0} + a_{0,1} = 0, a_{1,1} + a_{0,2} = 0 \Rightarrow a_{1,1} = 2$$

$j \backslash i$	0	1	2	3	...
0	$a_{0,0}$	0	1	0	0 ...
1	0	-2	0	0	
2	1	0	0		
3	0	0			

$$a_{3,0} = -a_{2,1}$$

$$a_{1,1} = -a_{0,2}$$

$$u(x, y) = a_{0,0} + a_{2,0} x^2 + a_{1,1} xy + a_{0,2} y^2$$

$$= a_{0,0} + x^2 - xy + y^2$$

$$u(x, 0) = a_{0,0} + x^2 \Rightarrow a_{0,0} = 0$$

$$\Rightarrow u(x, y) = x^2 - xy + y^2$$

$$a_{2,0} = a_{1,1}$$

Hmmmm neki je nerob...

$$u(x, y) = x^2 - 2xy + y^2$$

Pozabili smo

množititi $\geq i+1$ in

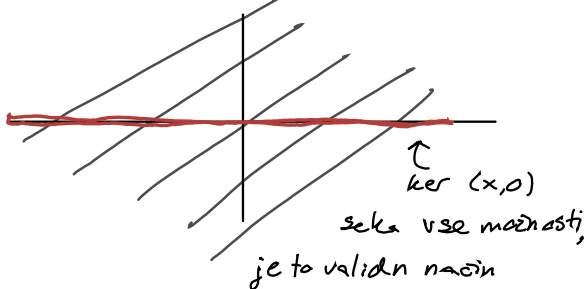
$j+1$ pri odvajanju

3. c)

$$u_x + u_y = 0 \quad u(x, 0) = x^2 \quad \leadsto u(x, y) = (x-y)^2$$

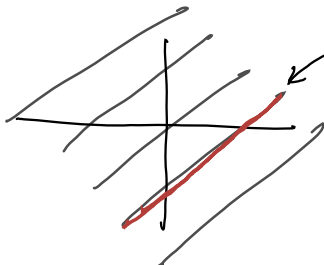
Naj bo $u(x, y) = g(x-y)$

$$\Rightarrow u_x + u_y = g'(x-y) - g'(x-y) = 0$$



Slab primer

$$u(x, x) = x^2$$



če imamo tak robni pogoji ne moremo dobiti rešitve

Ali obstaja, ali pa jure je veliko

$$u_{xy} + yu_y + u = 1$$

$$u_x(x, 0) = 1$$

$$u(x, x) = e^{-x^2}$$

$$yu_y + u = (yu)_y$$

$$u_{xy} + (yu)_y = 1 \leadsto u_x + yu = y + C(x)$$

0
||

$$v(x, 0): u_x(x, 0) + 0 \cdot 1 = 0 + C(x) \Rightarrow C(x) = 0$$

$$u_x + yu = y$$

homogeno:

$$u_x = -yu$$

$$\frac{du}{u} = -y dx$$

$$u = D(y)e^{-yx}$$

partikularna:

$$D_x(x, y)e^{-yx} = y$$

$\leadsto$

$$D(x, y) = \int y e^{yx} dx = e^{yx} + C(y)$$

$$u = 1 + C(y)e^{-yx}$$

$$u(x, x) = 1 + C(x)e^{-x^2} = e^{-x^2} \Rightarrow C(x) = \frac{e^{-x^2} - 1}{e^{-x^2}} = 1 - e^{x^2}$$

$$u(x, y) = 1 + (1 - e^{-y^2})e^{-yx}$$

$$u_{xy} + 2u_y = e^{-2x} \quad u(x,0) = e^{-x} \quad u(0,y) = 1$$

$$\hookrightarrow u_x(x,0) = -e^{-x}$$

integriramo po y

$$\Rightarrow u_x + 2u = ye^{2x} + c(x)$$

$$v(x,0): -e^{-x} + 2e^{-x} = c(x) = e^{-x}$$

$$u_x + 2u = ye^{-2x} + e^{-x}$$

hom:

$$\frac{du}{u} = -2dx \Rightarrow u = e^{-2x} c(y)$$

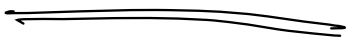
$$\text{part: } c_x(x,y)e^{-2x} = ye^{-2x} + e^{-x}$$

$$c(x,y) = \int (y + e^x) dx = yx + e^x + D(y)$$

$$u = yxe^{-2x} + e^{-x} + e^{-2x} D(y)$$

$$v(0,y): 0 + 1 + D(y) = 1 \Rightarrow D(y) = 0$$

$$\Rightarrow u = yxe^{-2x} + e^{-x}$$



KLE

4.3

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u)$$

$$S = \{ (x, y, z) \in \mathbb{R}^3 ; \quad z = u(x, y) \text{ u rešitev} \}$$

je integralne krivulje

Presek $S_1 \cap S_2$ je karakteristična krivulja

karakteristična krivulja $\gamma: \dot{\gamma}(t) = (a(\gamma(t)), b(\gamma(t)), c(\gamma(t)))$

Izrek:

- S_1, S_2 integralni ploskvi za isto KLE. $p \in S_1 \cap S_2 \Rightarrow$ karakteristična krivulja, ki vsebuje p leži celo v $S_1 \cap S_2$
- $S_1 \cap S_2$, če se ne sekata tangenčno, je karakteristična krivulja

Dokaz 2)

$$p \in C = S_1 \cap S_2$$

$$\pi_1 = T_p S_1 \quad \pi_2 = T_p S_2$$

$\dim(\pi_1 \cap \pi_2) = 1$, ker se ne sekata S_1 in S_2 tangenčno.

$\pi_1 \cap \pi_2$ tangenčna na C

$$a(p), b(p), c(p) \in \pi_1 \cap \pi_2 = T_p C$$

$$T_p C = \{ \lambda (a(p), b(p), c(p)) + p \ ; \ \lambda \in \mathbb{R} \}$$

C je tangenčna na karakteristično smer

$\Rightarrow C$ je karak. krivulja

①

$$xu_x + yu_y = 0 \quad (x, y, 0) = (a, b, c)$$

$$\dot{x} = x \rightsquigarrow x_0(s)e^s$$

$$\dot{y} = y \rightsquigarrow y_0(s)e^s$$

$$\dot{z} = 0 \rightsquigarrow z = z_0(s)$$

$$\varphi(t) = u(tx, ty)$$

$$\dot{\varphi}(t) = u_x \cdot x + u_y \cdot y = 0$$

$$\Rightarrow \varphi(t) = C(s) = C \quad \varphi(0) = u(0, 0)$$

$$\Rightarrow \varphi(t) = u(0, 0)$$

$$\varphi(1) = u(x, y) = C = u(0, 0)$$

$$\Rightarrow u(x, y) = u(0, 0)$$

$$② \quad u: \bar{D} \rightarrow \mathbb{R}$$

$$a(x,y) u_x + b(x,y) u_y = -u$$

$$x a(x,y) + y b(x,y) > 0 \quad \forall x,y \in \partial D$$

$$\min, \max D \Rightarrow u \equiv 0$$

$$\max D \Rightarrow u_x = u_y = 0 \quad \vee \quad (x_0, y_0)$$

$$\Rightarrow -u(x_0, y_0) = 0$$

$$\min D \Rightarrow -u(x_1, y_1) = 0 \Rightarrow \Rightarrow u \equiv 0$$

$$b) \quad (x, y, \lambda) \mapsto u(x, y) + \lambda(x^2 + y^2 - 1)$$

$$u_x(x, y) + 2\lambda x = 0$$

$$u_y + 2\lambda y = 0$$

$$x^2 + y^2 - 1 = 0$$

$$c) \quad t \mapsto u(t x_0, t y_0) \quad x_0, y_0 \text{ maks/min na robu}$$

$$\dot{f} = u_x x_0 + u_y y_0 \geq 0 \quad \vee \quad \text{dodici} \quad t=1$$

$$t=1: u_x(x_0, y_0) x_0 + u_y(x_0, y_0) y_0 = -2\lambda(x_0 + y_0)$$

$$\Rightarrow \lambda \leq 0$$

$$\text{Dokazimo da } u(x_0, y_0) > 0$$

$$a(x,y) (-2\lambda x_0) + b(x,y) (-2\lambda y_0) \stackrel{=-u}{< 0}$$

$$(-2\lambda) \underbrace{(a(x,y)x_0 + b(x,y)y_0)}_{> 0} < 0$$

$$\Rightarrow -2\lambda < 0 \Rightarrow \lambda > 0 \quad \times$$

$$\text{Stoga } u(x_0, y_0) < 0$$

$$\text{Kjer keli je minimum uvek kuli } \geq 0 \Rightarrow \Rightarrow$$

$$u \equiv 0$$

Metode karakteristik za enačbe 1. reda

enačba je

- $a(x, u(x)) \cdot \nabla u = b(x, u(x))$
- $u(x) = g(x)$ začetni pogoji
na krivulji Γ $g: \Gamma \rightarrow \mathbb{R}$

① Rešimo sistem za vse y po s

odkod $\begin{cases} \dot{z}(y, s) = b(x(y, s), z(y, s)) \\ \dot{x}(y, s) = a(x(y, s), z(y, s)) \end{cases}$

po s $x(y, 0) = y$; $z(y, 0) = g(y)$



② Najdemo $s(x)$, $y(x)$, potem $u(x) = z(y(x), s(x))$

$$\textcircled{1} \quad u(0, x) = g(x)$$

$$u_t + a \cdot \nabla u = 0$$

$$\dot{z}(y, s) = 0 \quad \leadsto \quad z = c(y)$$

$$\dot{t}(y, s) = 1 \quad \leadsto \quad t = s + c(y)$$

$$\dot{x}(y, s) = a \quad \leadsto \quad x = as + c(y)$$

$$(t(y, 0), x(y, 0)) = (0, y)$$

$$z(y, 0) = g(y)$$

$$z = c(y) = g(y)$$

$$t(y, 0) = c(y) = 0 \leadsto t(y, s) = s$$

$$x(y, 0) = c(y) = y \leadsto x(y, s) = as + y$$

$$y = x - at$$

$$u(x, y) = g(x - at)$$

b)

$$u_t + a \nabla u + u = f(x)$$

$$u_t + a \nabla u = f(x) - u$$

$$\dot{z}(y, s) = f(x(y, s)) - z(y, s)$$

$$\dot{t}(y, s) = 1 \leadsto t = s + C(y)$$

$$\dot{x}(y, s) = a \leadsto x = s + D(y)$$

$$x(y, 0) = g(y) \leadsto x = s + g(y)$$

$$t(y, 0) = 0 \leadsto t = s$$

$$\leadsto \dot{z}(y, s) = f(s + g(y)) - z$$

$$\text{hom: } \int \frac{dz}{z} = \int ds \quad \leadsto \quad z = C(y) e^{-s}$$

$$\text{par: } \dot{C}(y, s) e^{-s} = f(s + g(y))$$

$$\leadsto C = \int f(s + g(y)) e^s ds$$

$$z(y, s) = \int f(t + g(y)) e^t dt e^{-s}$$

$$= \int f(t) e^t dt e^{-g(y)} e^{-s}$$

$$u(x, y) = \left(\int f(t) e^t dt \right) e^{-x}$$

$$c) u_t + \nabla F(u) = 0$$

$$\begin{aligned} \nabla F(u(x_1, \dots, x_n)) &= (F'(u) \frac{\partial u}{\partial x_1}, \dots, F'(u) \frac{\partial u}{\partial x_n}) = \\ &= F'(u) \cdot \nabla u \end{aligned}$$

$$\dot{z} = 0 \quad \leadsto \quad z = c(y) = g(y)$$

$$\dot{x} = F'(z) \leadsto \dot{x} = F'(c) \leadsto x = F'(g(y))s + y$$

$$\dot{t} = 1 \quad \leadsto \quad t = s + D(y) \leadsto t = s$$

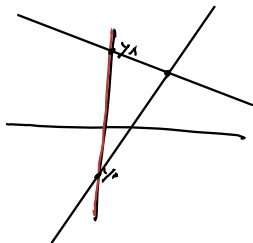
$$t(y, 0) = 0$$

$$x(y, 0) = y$$

$$z(y, 0) = g(y)$$

$$x(y, s) = F'(g(y))s + y$$

ke kaj se selekta



$$F'(g(y_0))s + y_0 = F'(g(y_1))s + y_0$$

če se selekta mora biti $g(\cancel{y_0}) = g(y_1)$

torej če se selekta, se ne more selekti
amrelet restar abetkeja buehno

$$a) u u_x + u_y = 1 \quad u(x, x) = \frac{x}{2}$$

$$a = u \quad \dot{x} = z \rightarrow x = \frac{1}{2}t^2 + \frac{s}{2}t + C(s) \rightarrow$$

$$b = 1 \quad \dot{y} = 1 \rightarrow t + C(s) \rightarrow y = t + s$$

$$c = 1 \quad \dot{z} = 1 \rightarrow t + C(s) \rightarrow z = t + \frac{s}{2}$$

$$X(s, 0) = S$$

$$Y(s, 0) = S$$

$$Z(s, 0) = \frac{s}{2}$$

$$x = \frac{1}{2}t^2 + \frac{1}{2}st + s \rightarrow s = -$$

$$t = y - s \rightarrow s = y - t$$

$$u(x, y) = \frac{s}{2} + t = \frac{s}{2} + y - s = y - \frac{s}{2}$$

$$x = \frac{1}{2}t^2 + \frac{1}{2}y t - t^2 + y - t = -\frac{1}{2}t^2 + \frac{1}{2}y t + y - t$$

$$\vdots \quad s = \frac{3x - y^2}{2 - y}$$

$$\in \mathbb{C} \quad x \neq 2 \quad \text{für } v \text{ reell}$$

$$a_2 \neq 2$$

$$b) -\sin(y) u_x + x u_y = 0 \quad u(x,0) = g(x) \quad x > 0$$

$$\begin{cases} \dot{x} = -\sin(y) \\ \dot{y} = x \\ \dot{z} = 0 \end{cases} \Rightarrow \begin{cases} \ddot{y} = \dot{x} = -\sin(y) \\ \frac{d^2}{dt^2} \sin(y) = \frac{d}{dt} \cos y \cdot \dot{y} = \\ = \dot{y} \cos y + \dot{y} (-\sin y) \dot{y} \end{cases}$$

$$x(s,0) = x$$

$$y(s,0) = 0$$

$$z(s,0) = g(s)$$

1. energija $E_1(x,y,z) = z$

$$\begin{aligned} \ddot{y} = \dot{x} = -\sin(y) \quad & x \dot{x} = x \ddot{y} = \dot{y} \ddot{y} = -x \sin(y) = -\dot{y} \sin y \\ 2. \text{ energija} \quad & x \dot{x} = (\cos y) \\ & \int x dx \int \cos y dt = \cos y + C \\ & \frac{1}{2} x^2 = \cos y + C \end{aligned}$$

$$E_2 = \frac{1}{2} x^2 - \cos y$$

Isčemo H .

$$E_1(x(0), y(0), z(0)) = g(s)$$

$$E_2(x(s,0), y(s,0), z(s,0)) = \frac{1}{2} s^2 - 1$$

$$H(a,b) = a - g(\sqrt{2b+1})$$

$$H(E_1, E_2) = g(s) - g(\sqrt{s^2}) = 0$$

$$H(E_1(x,y,z), E_2(x,y,z)) = 0 \quad \Leftrightarrow \quad \forall s, t, z$$

$$z - g(\sqrt{x^2 - 2\cos y + 2}) = 0$$

$$\Rightarrow u(x,y) = g(\sqrt{x^2 - 2\cos y + 2})$$

$$c) \quad x(y^2+u)u_x - y(x^2+u)u_y = (x^2-y^2)u$$

$$u(x,x) = C \in \mathbb{R} \quad x \neq 0$$

$$\dot{x} = x(y^2+z)$$

$$\dot{y} = -y(x^2+z)$$

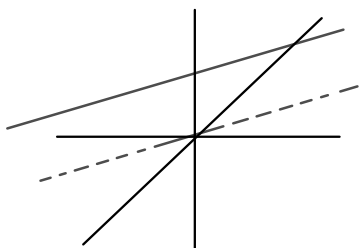
$$\dot{z} = z(x^2-y^2) = z(x-y)(x+y)$$

$$x(s,0) = s$$

$$y(s,0) = s$$

$$z(s,0) = c$$

$$\dot{x}(s,0) = s(s^2+c)$$



$$\dot{x}x + \dot{y}y = z(x^2-y^2) = \dot{z}$$

$$\frac{x^2}{2} + \frac{y^2}{2} = z + D(s)$$

$$E_1(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} - z$$

$$\frac{\dot{x}}{x} + \frac{\dot{y}}{y} + \frac{\dot{z}}{z} = y^2 - x^2 + x^2 - y^2 = 0$$

$$\ln x + \ln y + \ln z = C$$

$$xyz = F(s) \quad E_2(x,y,z) = xyz$$

$$E_1(s,0) = s^2 - c = a$$

$$E_2(s,0) = cs^2 = b \quad ca - b + c^2$$

$$H(a,b) = ca - b + c^2$$

$$\nabla H(E_1, E_2) =$$

$$c\left(\frac{x^2}{2} + \frac{y^2}{2} - z\right) - xyz + c^2 = 0$$

ist eine Lösung

$$z = u(x,y) = \frac{c^2 + \frac{c}{2}(x^2+y^2)}{c+xy}$$

Energija

Pri uporabi karakteristik v n -dimensionalnem sistemu, lahko iščemo n "energij". To so funkcije $E_j(x(t))$ za katere velja, da so ~~konstante~~

$$\frac{d}{dt} E_j(x(t)) = 0$$

Nato predpišemo $H(E_1, \dots, E_n) = 0$ (oz poljubno konstanto) in z uporabo začetnih pogojev določimo H , ki zadošča $H(E_i(x(0)), \dots) = 0$

$$u_x + xu_y = u$$

$$\dot{x} = 1$$

$$\dot{y} = x$$

$$\dot{z} = z$$

$$u(1, y) = h(y) \quad h \in \mathcal{C}(\mathbb{R})$$

$$x(s, 0) = 1$$

$$y(s, 0) = s$$

$$z(s, 0) = h(y)$$

$$x = t + 1$$

$$y = \frac{1}{2}t^2 + t + s$$

$$z = h(y) e^{x-1}$$

$$\dot{z} = z$$

$$\frac{dz}{z} = dt$$

$$z = ce^t = h(y) e^{x-1}$$

$$t = x - 1$$

$$s = y - \frac{1}{2}(x-1)^2 - (x-1)$$

$$u(x, y) = h\left(y - \frac{1}{2}(x-1)^2 - (x-1)\right) e^{x-1}$$

$$x^2 u_x + y^2 u_y = (x+y)u \quad u(x,1)=1$$

$$x(3,0)=6$$

$$\dot{x} = x^2 \leadsto -\frac{1}{x} = t + C \leadsto x = \frac{-1}{t+C} \leadsto x = \frac{s}{1-ts}$$

$$\dot{y} = y^2 \leadsto y = \dots = \frac{1}{1-t}$$

$$\dot{z} = (x+y)z$$

$$\frac{\dot{z}}{z} = \frac{1}{1-t} + \frac{s}{1-ts}$$

$$\ln z = -\ln(1-t) - \ln(1-ts) +$$

należy pisać
sę za zmienną
wzrost

$$z = \frac{1}{1-t} \cdot \frac{1}{(1-t)(1-ts)}$$

$$z = y \cdot \frac{x}{s}$$

$$s = x(1-ts)$$

$$s(1+tx) = x \leadsto s = \frac{x}{1+tx}$$

$$tx = (1-t)$$

$$(1-t)y = 1$$

$$t = -\frac{1}{y} + 1 = 1 - \frac{1}{y} = \frac{y-1}{y}$$

$$s = \frac{x}{1 - \frac{(y-1)x}{y}} = \frac{xy}{y+x-xy}$$

$$z = y + x - xy \leadsto u(x,y) = y + x - xy$$

$$u(x,0) = 1$$

$$x = \frac{1}{1+c}$$

$$y = \frac{1}{1+c}$$

$$x(s,0) = s$$

$$y(s,0) = 0$$

$$z(s,0) = 1$$

$$\frac{1}{c} = s \Rightarrow \frac{1}{c} = \frac{1}{s} \Rightarrow x = \frac{s}{1+s}$$

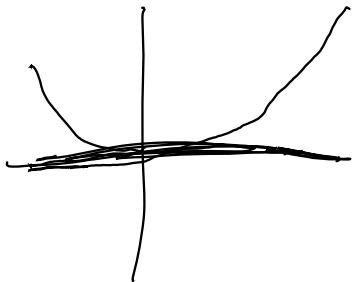
$$\frac{1}{c} = 0 \Rightarrow c = \infty \quad y = 0$$

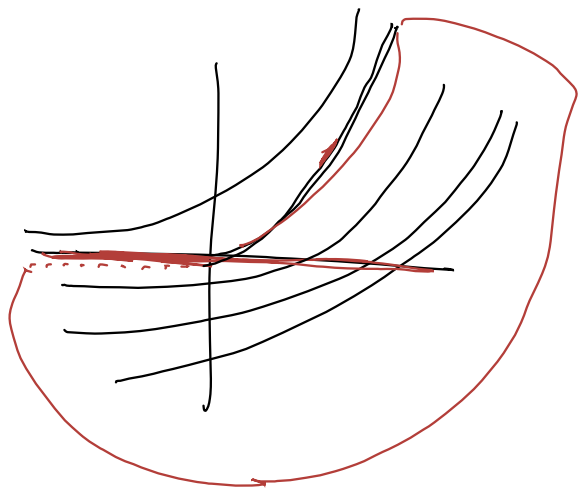
$$\dot{z} = \frac{s}{1+s} z$$

$$\ln z = -\ln(1+s) + C \Rightarrow$$

$$z = \frac{E(s)}{1+s} = \frac{1}{1+s} \quad \text{ne marela no izraz}$$

$\Rightarrow x \text{ in } y \Rightarrow \text{ne drzaga cuclu}$





ogrinjača družine ravnin skozi točko

Naj bo $\Pi_\lambda = \{ (x, y, z) \mid z = \psi(x, y, \lambda) = a(\lambda)x + b(\lambda)y \}$
 družina ravnin skozi izhodišče
 $a, b \in \mathcal{C}(I)$ $I \subseteq \mathbb{R}$ interval, da je
 $\vec{n}(\lambda) = (a(\lambda), b(\lambda), -1)$ injektivna
 in $w(a', b') = \begin{vmatrix} a' & b' \\ a'' & b'' \end{vmatrix} \neq 0$

Opazimo: $z = -1$ komponenta je normalizacija,
 ki da $\Pi_{\lambda_1} = \Pi_{\lambda_2} \Leftrightarrow \lambda_1 = \lambda_2$

Cfj: Najdemo $\sigma(x, y)$, ki definira $\sigma = \{ (x, y, \psi(x, y)) \}$
 da velja

- $\sigma \subseteq \bigcup \Pi_\lambda$
- $p_0 = (x_0, y_0, z_0) \in \sigma \cap \Pi_\lambda$ za nek λ
 $\Rightarrow$ normalni vektor za σ v točki p_0
 je sorazmeren z $\vec{n}(\lambda)$

Opazka: $\sigma = \Pi_\lambda$ za fiksni λ je trivialna
 rešitev, ki jo ignoriramo

ker je $\vec{n}(\lambda)$ injektivna, če $p_0 \in \sigma \cap \Pi_\lambda$ in $p_0 \in \sigma \cap \Pi_\mu$
 potem $\lambda = \mu$

Consider the map $P: \mathbb{R}^3 \rightarrow \mathbb{R}^2$
 če $(x, y, z) = P(x, y, z)$ za nek $(x, y, z) P \in \sigma$
 potem $p \in \Pi_\lambda$ za nek unikatni $\lambda = \lambda(x, y)$

$$\lambda: P(\sigma) \rightarrow \mathbb{R}$$

$$(x, y) \mapsto \lambda(x, y)$$

ker $\sigma = \{ z = \psi(x, y) \} \subseteq \bigcup \Pi_\lambda$ imamo
 $\psi(x, y) = a(\lambda(x, y))x + b(\lambda(x, y))y = \psi(x, y, \lambda(x, y))$

normala za σ je paralelna $\vec{n}(\lambda)$, torej
 parametrizacija σ je $u(x, y) = (x, y, \psi(x, y))$
 normala je podana z $u_x \times u_y$
 računamo:

$$u_x = (1, 0, \psi_x(x, y)) = \psi_x(x, y, \lambda(x, y)) + \psi_\lambda(x, y, \lambda(x, y))\lambda_x(x, y)$$

$$u_y = (0, 1, \psi_y(x, y)) = \psi_y(x, y, \lambda(x, y)) + \psi_\lambda(x, y, \lambda(x, y))\lambda_y(x, y)$$

$$-u_x \times u_y = (\psi_x + \psi_\lambda \lambda_x, \psi_y + \psi_\lambda \lambda_y, -1)$$

$$= (\psi_x, \psi_y, -1) + \psi_\lambda (\lambda_x, \lambda_y, 0)$$

Spominimo se $\psi = a(x)x + b(y)y$

$$= \underbrace{a(\lambda(x, y)), b(\lambda(x, y)), -1}_{\vec{n}(\lambda(x, y))} + \psi_\lambda'(\lambda_x, \lambda_y, 0)$$

Želimo da je $\leftarrow = 0$

1. $(\lambda_x, \lambda_y, 0) = 0 \Rightarrow \nabla \lambda = 0 \Rightarrow \lambda$ konstantna
 $\Rightarrow \psi(x, y) = a(\lambda)x + b(\lambda)y \Rightarrow \sigma \subseteq \Pi_\lambda$ trivialna
 rešitev

2. $\psi_\lambda(x, y, \lambda(x, y)) = 0 \quad \forall x, y \in P(\sigma)$

vseeno mi je, kaj je } pri hitrosti

a)

$$(u_x u_y)^2 = u \quad u(x, 0) = x$$

$$(p_z)^2 = z$$

$$F(x, y, z, p, q) = (p_z)^2 - z$$

$$\dot{x} = \frac{\partial F}{\partial p} = 2(p_z)q$$

$$\dot{y} = 2(p_z)p$$

$$\dot{z} = 2p^2 q^2 + 2p^2 q^2 = 4p^2 q^2$$

$$\dot{p} = -(-p) = p$$

$$\dot{q} = q$$

$$p = C(s) e^t$$

$$q = D e^t$$

$$\dot{z} = 4 \dot{z} \Rightarrow z = E(s) e^{4t}$$

$$\dot{z} = 4 e^{4t} C D$$

$$z = e^{4t} (C D)^2 = (e^{4t} - 1) C(s)^2 D(s)^2 + E(s)$$

$$\dot{x} = 2 C^2 D e^{3t} \Rightarrow x = \frac{2}{3} C^2 D (e^{3t} - 1) + F(s)$$

$$\dot{y} = \frac{2}{3} C D^2 e^{3t} \Rightarrow y = \frac{2}{3} C D^2 (e^{3t} - 1) + G(s)$$

$$u(x, 0) = x$$

$$x(s, 0) = s = F(s)$$

$$y(s, 0) = 0 = G(s)$$

$$z(s, 0) = 0 = E(s)$$

$$p(s, 0) = u_x(x(s, 0), y(s, 0)) = 1 \Rightarrow C(s) = 1$$

$$q(s, 0) = \pm \sqrt{s}$$

$$(u_x u_y)^2(x, 0) = u(x, 0) = s$$

$$(1 \cdot u_y(x, 0))^2 = x$$

$$q_0 = u_y(x, 0) = \pm \sqrt{x}$$

$$u(x, 0) = \pm \sqrt{x} y + C(x)$$

$$z = (e^{4t} - 1)s + s$$

$$x = \frac{2}{3}(e^{3t} - 1)s + s \quad s = \frac{x}{1 + \frac{2}{3}(e^{3t} - 1)}$$

$$y = \frac{2}{3}(e^{3t} - 1)\sqrt{s}$$

$$p = e^t$$

$$e^{3t} = \frac{3}{2} \frac{y}{\sqrt{s}} + 1$$

$$q = e^t \sqrt{s}$$

$$s = \frac{x}{1 + \frac{y}{\sqrt{s}}}$$

$$s + \sqrt{s} y = x$$

$$s = \left(\frac{y}{\frac{2}{3}(e^{3t} - 1)} \right)^2$$

$$x = \frac{y^2}{\frac{2}{3}(e^{3t} - 1)} + \left(\frac{y}{\frac{2}{3}(e^{3t} - 1)} \right)^2$$

$$s = \left(\frac{x^2}{1 \pm \sqrt{y^2 - 4x}} \right)^2$$

$$v^2 x - v y^2 - y^2 = 0$$

$$D = \sqrt{y^4 - 4xy^2} = y \sqrt{y^2 - 4x}$$

$$\frac{2}{3}(e^{3t} - 1) = \frac{y^2 \pm y \sqrt{y^2 - 4x}}{x^2}$$

$$u = \frac{3}{2} \left(\frac{y^2 \pm y \sqrt{y^2 - 4x}}{x^2} \right) \left(\frac{x^2}{1 \pm \sqrt{y^2 - 4x}} \right)^2 + \left(\frac{x^2}{1 \pm \sqrt{y^2 - 4x}} \right)^2$$

$$b) u_x^2 + u_y = yu$$

$$u(x,0) = x$$

$$F(x,y,z,p,q) = p^2 + z - yz$$

$$\dot{x} = 2p$$

$$x(s,0) = s$$

$$\dot{y} = 1$$

$$y(s,0) = 0$$

$$\dot{z} = p \cdot (2p) + z - 1$$

$$z(s,0) = s$$

$$\dot{p} = -0 + py = py$$

$$p(s,0) = 1$$

$$\dot{q} = z + zy$$

$$q(s,0) = -1$$

$$y = t + c(s) = t$$

$$dp = p dt \Rightarrow \ln p = \frac{1}{2} t^2 + c \Rightarrow p = p_0 e^{\frac{1}{2} t^2} = e^{\frac{1}{2} t^2}$$

$$dx = 2e^{\frac{1}{2} t^2} dt \Rightarrow x =$$

$$2 \int e^{\frac{1}{2} t^2} dt = \int \sqrt{2} u^{-\frac{1}{2}} e^u du$$

$$u = \frac{1}{2} t^2$$

$$du = t dt = \sqrt{u} dt$$

$$\dot{z} = 2e^{t^2} + z$$

$$\dot{q} = z + q$$

$$\ddot{z} = \ddot{z} + z + t\dot{z} = 2e^{t^2} z$$

$$\ddot{q} + t\dot{q} = 2e^{t^2}$$

$$\text{hom: } \frac{d\dot{q}}{\dot{q}} = -t dt \Rightarrow \dot{q} = A(t) e^{-t}$$

$$\text{part: } A' e^{-t} = 2e^{t^2}$$

$$A' = 2e^{t^2+t}$$

$$b) u_x^2 + u_y = yu$$

$$u(x,0) = x$$

$$F(x,y,z,p,q) = p^2 + q - yz$$

$$\dot{x} = 2p$$

$$x(s,0) = s$$

$$\dot{y} = 1$$

$$y(s,0) = 0$$

$$\dot{z} = p \cdot (2p) + q \cdot 1$$

$$z(s,0) = s$$

$$\dot{p} = -0 + py = py$$

$$p(s,0) = 1$$

$$\dot{q} = z + qy$$

$$q(s,0) = -1$$

Linearne PDE 2. reda v $\mathbb{R}^2$

$$\underline{a(x,y)u_{xx} + 2b(x,y)u_{xy} + c(x,y)u_{yy} + d(x,y)u_x + e(x,y)u_y + f(x,y)u + g(x,y) = 0}$$

$$\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix} u$$

$$\Delta = b^2 - ac = -\det \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

kanonična oblika

$\Delta > 0$ hiperbolični tip

oziramo
 $u_{xy} + \text{nižji red} = 0$
 $u_{xx} - u_{yy} + \text{nižji red} = 0$

$\Delta = 0$ parabolični tip

$$u_{xx} + \text{nižji red} = 0$$

$\Delta < 0$ eliptični tip

$$u_{xx} + u_{yy} + \text{nižji red} = 0$$

Hiperbolični tip:

$$\frac{\partial y}{\partial x} = \frac{b \pm \sqrt{\Delta}}{a} \leadsto y = f_{\pm}(x, c)$$

Novi koordinati $\xi(x, y), \eta(x, y)$

velja $y = f_+(x; \xi(x, y)), y = f_-(x; \eta(x, y))$

Primer: $\frac{\partial y}{\partial x} = 2 \pm 1 \leadsto y = \begin{cases} 3x + c = f_+ \\ x + c = f_- \end{cases}$

$$\xi(x, y) = y - 3x$$

$$\eta(x, y) = y - x$$

• Če $a=0 \leadsto \frac{\partial x}{\partial y} = \frac{b \pm \sqrt{\Delta}}{c}$

• Če $a=0$ in $c=0 \leadsto$ enačba je že v kanonični obliki

Parabolični tip $\Delta = 0$

$$\frac{dy}{dx} = \frac{b}{a} \quad (= \frac{b \pm \sqrt{\Delta}}{a}) \Rightarrow y = f(x, c)$$
$$y = f(x, \xi(x, y))$$

$\eta(x, y)$ je poljubna (amperke še koordinati)

Elipson: tip $\Delta < 0$

$$\frac{dy}{dx} = \frac{b \pm i\sqrt{-\Delta}}{a} = \frac{b \pm \sqrt{\Delta}}{a} \Rightarrow y = f(x, c)$$

Iščemo $\Phi(x, y)$: $y = f(x, \Phi(x, y))$

$$\xi = \operatorname{Re} \Phi \quad \eta = \operatorname{Im} \Phi$$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Diagonalisieren

$$(\lambda I - A) = \begin{bmatrix} \lambda - 1 & 0 & -1 \\ 0 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 1 \end{bmatrix} =$$

$$= (\lambda - 1)((\lambda - 1)^2 - 1) = (\lambda - 1)(\lambda^2 - 2\lambda) = \lambda(\lambda - 1)(\lambda - 2)$$

$$\lambda = 0: e_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda = 1: e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 2: e_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 2 \end{bmatrix}$$

$$A = Q^T D Q$$

$$z_1 = x - z$$

$$z_2 = y$$

$$z_3 = x + z$$

$$z = Q \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$u = u(x(z), y(z), z(z)) = v(z)$$

$$x = \frac{1}{2}(z_1 + z_3)$$

$$u_x = (z_1)_x v_{z_1} + (z_2)_x v_{z_2} + (z_3)_x v_{z_3}$$

$$y = z_2$$

$$z = \frac{1}{2}(z_3 - z_1)$$

...

$$u_x = v_{z_1} + v_{z_3}$$

$$\partial_x = \partial_{z_1} + \partial_{z_3}$$

$$u_y = v_{z_2}$$

$$\partial_y = \partial_{z_2}$$

$$u_z = -v_{z_1} + v_{z_3}$$

$$\partial_z = \partial_{z_3} - \partial_{z_1}$$

$$u_{xx} = v_{z_1} + 2v_{z_1 z_3} + v_{z_3}$$

$$u_{yy} = v_{z_2 z_2}$$

$$u_{zz} = v_{z_1} - 2v_{z_1 z_3} + v_{z_3 z_3}$$

$$u_{xz} = -v_{z_1 z_1} + \dots$$

$$u_{xx} + 2u_{xz} + u_{yy} + u_{zz} = 0$$

$\Rightarrow$

$$4v_{z_2} + v_{z_1} = 0$$

n: konstante
abw.

Normalisierte Q

$$\tilde{z}_2 = \frac{z_2}{2} \Rightarrow \tilde{v}_{z_2} = \frac{z_2}{2} \quad v(z_1, z_2, z_3)$$

$$= v(z_1, \frac{z_2}{2}, z_3)$$

$$\tilde{v}_2 = v_2 \cdot \frac{1}{2}$$

$$\tilde{v}_{z_2} = v_{z_2} \cdot \frac{1}{2}$$

$$\Rightarrow \tilde{v}_{z_2} + v_{z_1} = 0$$

D'Alembertova formula na $[0, \infty) \times \mathbb{R}$

$$u(t, x) = \frac{1}{2} (f(x-t) + f(x+t)) + \frac{1}{2} \int_{x-t}^{x+t} g(\xi) d\xi + \frac{1}{2} \iint_{\Delta} F(\tau, \xi) d\tau d\xi$$

gde je $\Delta(t, x) = \{(\tau, \xi) \in \mathbb{R}^2 \mid 0 \leq \tau \leq t, |\xi - x| \leq t - \tau\}$

$$u_{tt} - u_{xx} = F(t, x)$$

$$u(0, x) = f(x)$$

$$u_t(0, x) = g(x)$$

$$u_{tt} - u_{xx} = 0 \quad \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

$$u(0, x) = u_t(0, x) = 0$$

$$k, \omega > 0$$

$$u_{tt} - u_{xx} = \sin(kx) \cos(\omega t)$$

$$u(0, x) = u_t(0, x) = 0$$

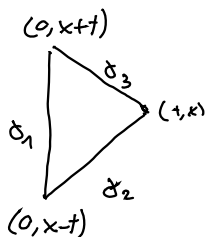
$$f(x) = g(x) = 0$$

$$F(x, t) = \sin(kx) \cos(\omega t)$$

$$u(t, x) = \frac{1}{2} \iint_{\Delta} F(\tau, z) d\tau dz$$

$$\Delta = \{(\tau, z) \in \mathbb{R}^2 \mid 0 \leq \tau \leq t, |z - x| \leq t - \tau\}$$

$$-\iint_D (y_x - x_y) dx dy = \int_{\partial D} (x, y) ds$$



$$X = -\frac{1}{\omega} \cos(kx) \cos(\omega t)$$

$$Y = 0$$

$$X_y = \sin(kx) \cos(\omega t)$$

$$\iint_{\Delta} \dots = - \iint_{\Delta} \left(-\frac{1}{\omega} \cos(kx) \cos(\omega t), 0 \right) ds$$

$$\gamma_1 : 0$$

$$\gamma_2 : \gamma_2(s) = (s, x - (t-s)) \quad s \in (0, t)$$

$$\gamma_3 = (t-s, x+s) \quad s \in (0, t)$$

$$\int_0^t \left(\frac{1}{k} \cos(\omega s) \cos(k(x-t+s)), 0 \right) \cdot (1, 1) ds =$$

$$= \int_0^t \frac{1}{2} (\cos(\omega s + k(x-t+s)) + \cos(\omega s - k(x-t+s))) ds =$$

$$= \int_0^t \frac{1}{2} (\cos(s(\omega + k) + x - t) + \cos(s(\omega - k) - x + t)) ds$$

$$u_{tt} - u_{xx} = 0 \quad \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

$$u(0, x) = u_t(0, x) = 0$$

d'Alembert's formula

ze vellet

$$u(x, t) = 0 + \frac{1}{2} \int_{-1}^1 1 dz + 0 = 1$$

$$\lim_{t \rightarrow \infty} = 1$$

$$u(x, t) = \frac{f(x+t) + f(x-t)}{2}$$

$$\sup = ?$$

$$u(x, t) \leq \frac{1}{2}(1+1) + \frac{1}{2} \cdot 2 + 0 = 2$$

$$x+t = m < 1 \quad x > 0 \Rightarrow t < 1$$

$$u(x, t) = \frac{1}{2}(1+1) + \frac{1}{2} \int_{x-t}^{x+t} g(z) dz$$

$$x=0 \quad t=m < 1$$

$$u(x, t) = 1 + \frac{1}{2} \int_{-m}^m g(z) dz = 1 + m$$

$$\lim_{m \rightarrow 1} u(x, t) = 1 \Rightarrow \sup = 2$$

$$u_{tt} - u_{xx} = 0$$

$$g(x) = 0 \quad t, x \in [0, \infty]$$

$$u(t, 0) = h(t)$$

$$u(t, 0) = h(t)$$

$$u(x, t) = \frac{f(x-t) + f(x+t)}{2}$$

$$u(t, 0) = \frac{1}{2} (f(t) + f(-t)) = h(t)$$

??

$$\begin{aligned}
 u_{tt} - u_{xx} &= 0 & u(0, x) &= f(x) \\
 u_t(0, x) &= 0 & u(t, 0) &= h(t)
 \end{aligned}$$

$$\delta = 0 + 1 = 1$$

$$\frac{dx}{dt} = \frac{0 \pm \sqrt{1}}{1} = \pm 1$$

$$x = \pm t + C$$

$$\zeta(t, x) = x + t$$

$$\eta(t, x) = x - t$$

$$w_{\zeta\eta} = 0$$

$$w_{\zeta} = C(\zeta)$$

$$w = C(\zeta) + D(\eta)$$

$$u = \underset{F}{C(x+t)} + \underset{G}{D(x-t)}$$

$$x, t > 0$$

$$u(0, x) = F(x) + G(x) = f(x)$$

$$u_t(t, 0) = F'(t) - G'(t) = h(t)$$

$$u_t(0, x) = F'(x) - G'(x) = 0$$

$$F'(x) = G'(x) \Rightarrow F(x) = G(x) + C$$

$$G(x) = \frac{f(x) - C}{2}$$

$$F(x) = \frac{f(x) + C}{2}$$

$$F(x) + G(x) = \frac{2f(x)}{2} \Rightarrow C = 0 \quad \text{for } x > 0$$

$$\text{za } x > 0 \quad \text{vemo}$$

$$F(x) = \frac{1}{2} f(x)$$

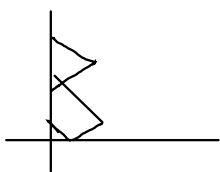
$$G(x) = \frac{1}{2} f(x)$$

$$F(x) = h(x) - G(-x) = \frac{1}{2} f(x)$$

$$\Rightarrow G(x) = h(-x) - \frac{1}{2} f(-x) \quad \text{za } x < 0$$

Splorna rešitev:

$$u(x, y) = \begin{cases} \frac{1}{2}(f(x+t) + f(x-t)) & ; x-t > 0 \\ \frac{1}{2}f(x+t) - \frac{1}{2}f(-x+t) + h(-x+t) & ; x-t < 0 \end{cases}$$



$$u_{tt} - u_{xx} = 0 \quad t > 0 \quad x \in \mathbb{R}$$

$$u(0, x) = x$$

$$u_x(0, x) = 0$$

$$\text{is also } w, \text{ de } b_0 \quad w_{tt} - w_{xx} = 0$$

$$w = u + \frac{1}{12}x^4 - x$$

$$w_t = u_t$$

$$w_x = u_x + \frac{1}{3}x^3 - 1$$

$$w_{tt} = u_{tt}$$

$$w_{xx} = u_{xx} + x^2$$

$$w(0, x) = u(0, x) + \frac{1}{2}x^4 - x = \frac{1}{2}x^4$$

$$w_t(0, x) = u_t(0, x) = 0$$

$$w = F(x+t) + G(x-t)$$

$$w(x, 0) = F(x) + G(x) = \frac{1}{2}x^4$$

$$w_t(0, x) = F'(x) - G'(x) = 0$$

$$\Rightarrow F(x) = G(x) + C$$

$$2G(x) + C = \frac{1}{2}x^4$$

$$G(x) = \frac{1}{4}x^4 - \frac{1}{2}C$$

$$F(x) = \frac{1}{4}x^4 + \frac{1}{2}C$$

$$w(x, t) = \frac{1}{4}((x+t)^4 + (x-t)^4)$$

$$u(x, t) = \frac{1}{4}((x+t)^4 + (x-t)^4) - \frac{1}{12}x^4 + x$$

$$u, v \text{ real}; \quad u_{tt} - u_{xx} = 0 \quad t > 0$$

$$u_x(t, a) = u_x(t, b) = 0$$

$$u_t v_t + u_x v_x \stackrel{?}{=} F(x+t) + G(x-t)$$

$$u = f(x+t) + g(x-t)$$

$$v = h(x+t) + k(x-t)$$

$$\begin{aligned} u_t v_t + u_x v_x &= (f'(x+t) - g'(x-t))(h'(x+t) - k'(x-t)) \\ &\quad + (f'(x+t) + g'(x-t))(h'(x+t) + k'(x-t)) = \\ &= \underbrace{2f'(x+t)h'(x+t)}_{F(x+t)} + \underbrace{2g'(x-t)k'(x-t)}_{G(x-t)} \end{aligned}$$

$$F \stackrel{?}{=} t \mapsto \int_a^b u_t v_t + u_x v_x$$

$$\begin{aligned} F &\stackrel{?}{=} \int_a^b \frac{\partial}{\partial t} (u_{tt} v_t + u_t v_{tt} + u_{xt} v_x + u_x v_{xt}) dx = \\ &= \int_a^b (u_{tt} v_t + u_t v_{tt}) dx + \underbrace{u_t v_x \Big|_a^b}_0 - \int_a^b (u_t v_{xx} + u_{xx} v_t) dx \end{aligned}$$

$$= \int_a^b v_t (\underbrace{u_{tt} - u_{xx}}_0) + u_t (\underbrace{v_{tt} - v_{xx}}_0) dx = 0$$

$$u_x^2 u_{xx} \pm 2u_x u_y u_{xy} + u_y^2 u_{yy} = 0$$

$$\underbrace{\left(\frac{1}{3} u_x^3\right)_x}_{\left(\frac{1}{3} u_x^3\right)_x} \quad \underbrace{\left(\frac{1}{3} u_y^3\right)_y}_{\left(\frac{1}{3} u_y^3\right)_y}$$

$$u = f(x)g(y)$$

$$u_x = f'(x)g(y)$$

$$u_y = f(x)g'(y)$$

$$u_{xx} = f''(x)g(y)$$

$$u_{xy} = f'(x)g''(y) \quad u_{yy} = f(x)g''(y)$$

$$\Rightarrow f'(x)^2 g(y)^3 f'''(x) \pm 2f'(x)^2 g(y) f(x) g''(y) + g'(y)^2 f^3(x) g''(y)$$

$$\text{delimoz } f^3 g^3$$

$\Rightarrow$

$$\frac{f^2 f'''}{f^3} \pm \frac{2f'(x)^2 g'(y)}{g^2(y) f^2(x)} + \frac{g'^2 g'(y)}{g^3(y)} =$$

$$\text{Nehmen wir } f(x) = e^{\lambda x} \quad \Rightarrow \lambda e^x \sim x^2 e^x$$

$$g(y) = e^{\mu y}$$

$$\lambda^4 \pm 2\lambda^2 \mu^2 + \mu^4 = 0$$

$$\Rightarrow (\lambda^2 \pm \mu^2)^2 = 0$$

$$\therefore (\lambda - \mu)(\lambda + \mu)(\lambda^2 + \mu^2) = 0 \quad \Rightarrow \lambda = \pm \mu$$

$$+ \therefore \lambda^2 + \mu^2 = 0 \quad \Rightarrow \lambda = 0 \wedge \mu = 0$$

$$u) \quad u = e^{\lambda x} e^{\pm \lambda x} = e^{\lambda(x \pm y)}$$

$$u_{tt} - u_{xx} = 0 \quad t \in (0, \infty)$$

$$u(0, x) = f(x) \quad x \in (0, 1)$$

$$\mu_+(0, x) = \infty$$

$$u(t, 0) = u(t, 1) = 0$$

$$u(x,t) = F(x-t) + G(x+t)$$

$$u(0, x) = f(x) + g(x) = f(x)$$

$$u_+(0, x) = -f'(x) + G'(x) = 0$$

$$u(t, 0) = F(-1) + g(t) = 0$$

$$u(t, 1) = F(1-t) + G(1+t) = 0$$

$$F(x) = G(x) + C$$

$$F(x) = \frac{f(x) - c}{2}$$

$$B(x) = \frac{f(x) + C}{2}$$

$$F(x) = \frac{f(x)}{2}$$
$$G(x) = \frac{f(x)}{2}$$

to je za $x \in (0, 1)$

$$-F(-t) = G(t) \leadsto f(-t) = -f(t) \quad \forall t \in (0,1)$$

$$F(1-t) = -G(1+t)$$

$$t \in (0, 1) \rightsquigarrow$$

$$f(1+t) = -G(1+t)$$

$$G(t) = f(2 - \frac{1}{2}t) \quad \text{for } t \in (1, 2)$$

$$t \in (n, n+1) \quad \leadsto x \in (n, n+1)$$

$$G(1+t) = -F(1-t)$$

$$G(X) = -F(2-X)$$

$$G(x) = \begin{cases} -F(2-x) & ; x \geq 1 \\ \frac{1}{2}f(x) & ; x \in (0,1) \\ -\frac{1}{2}f(-x) & ; x < 0 \end{cases}$$

$$t \in (0, 2) \Rightarrow F(1-t) = -G(1+t)$$

$$s \in (1, 3) \Rightarrow F(s) = -G(2-s) = -f(2-s) \cdot \frac{1}{2}$$

$$F(x) = \begin{cases} -\frac{1}{2}f(-x) & ; x \in (-1, 0) \\ \frac{1}{2}f(x) & ; x \in (0, 1) \\ -\frac{1}{2}f(2-x) & ; x \in (1, 2) \\ \frac{1}{2}f(2+x) & ; x \in (2, 3) \end{cases}$$

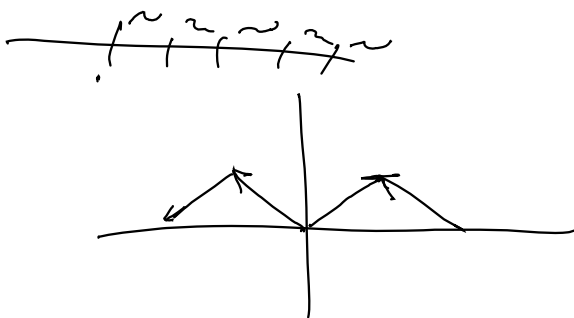
$$S = 1 - t$$

$$t \in (-n-2, n) \quad F(1+t) = -G(1+t)$$

$$\Rightarrow s \in (n-1, n+1) \quad F(s) = -G(2-s) = F(s-2)$$

$$SE(2n, 2n+1) \leadsto F(s) = F(s-2) = F(s-2n) = \frac{1}{2} f(s-2n)$$

$$s \in (2n-1, 2n) \leadsto F(s) = F(s-2) = F(s-2n) = -\frac{1}{2} f(2n-s)$$



$$u_t - k u_{xx} = 0 \quad \begin{array}{l} t \in (0, \infty) \\ x \in [0, 2\pi] \end{array}$$

$$u(t, 0) = u(t, 2\pi)$$

$$u_x(t, 0) = u_x(t, 2\pi)$$

$$u(0, x) = f(x)$$

$$u(x, t) = F(x - \sqrt{k}t) + G(x + \sqrt{k}t) \quad c = \sqrt{k}$$

$$u(0, x) = F(x) + G(x) = f(x)$$

$$u_x = F' + G'$$

$$F'(-ct) + G'(ct) = F'(2\pi - ct) + G'(2\pi + ct)$$

$$-F'(-ct) + G'(ct) = -F'(2\pi - ct) + G'(2\pi + ct) + D$$

$$u_t - u_{xx} = 0 \quad (t, x) \in (0, \infty) \times I \quad I \subset \mathbb{R}$$

z robnimi in začetnimi pogoji

Pišemo: $u(t, x) = v(t) w(x)$

$$\Rightarrow v'(t) w(x) = v(t) w''(x)$$

$$\Rightarrow \frac{v'}{v}(t) = \frac{w''}{w}(x) \equiv \lambda \in \mathbb{R}$$

$$\Rightarrow v(t) = c e^{\lambda t}$$

$$w(x) = \alpha e^{\sqrt{\lambda} x} + \beta e^{-\sqrt{\lambda} x} \quad \alpha, \beta \in \mathbb{C}$$

rešba je linearna $\Rightarrow$

$$u(t, x) = \sum_{k=0}^{\infty} v_k(t) w_k(x)$$

Odločimo $w_k(x) = \alpha_k e^{\sqrt{\lambda_k} x} + \beta_k e^{-\sqrt{\lambda_k} x}$, $v_k = c_k e^{\lambda_k t}$

Določimo λ_k odvisno od robnega pogoja
in c_k odvisno od začetnega pogoja

$$u(t, a) = u(t, b) = 0$$

$$V = \left\{ w \in L^2([a, b], \mathbb{R}) ; w(a) = w(b) = 0 \right\}$$

$$u_t - k u_{xx} = 0 \quad t \in (0, \infty) \\ x \in [0, 2\pi)$$

$$u(t, 0) = u(t, 2\pi)$$

$$u_x(t, 0) = u_x(t, 2\pi)$$

$$u(0, x) = f(x) = \sin x$$

$$u(t, x) = \sum v_k(t) w_k(x)$$

$$u(t, 0) = u(t, 2\pi) \Rightarrow \sum v_k(t) w_k(0) = \sum v_k(t) w_k(2\pi) \\ \Rightarrow \sum v_k(t) (w_k(0) - w_k(2\pi)) = 0$$

$$u_x(t, x) = \sum v_k(t) w'_k(x) \Rightarrow \\ \sum v_k(t) (w'_k(0) - w'_k(2\pi)) = 0$$

$$u(0, x) = f(x) \Rightarrow \sum v_k(0) w_k(x) = f(x)$$

$$v_k = c_k e^{\lambda_k t} \leadsto \sum c_k e^{\lambda_k t} (w_k(0) - w_k(2\pi)) = 0$$

$$w_k(0) = \alpha_k + \beta_k = 0$$

$$w_k(\pi) = \alpha_k e^{\sqrt{\lambda_k} \pi} + \beta_k e^{-\sqrt{\lambda_k} \pi} = 0$$

$$e^{\sqrt{\lambda_k} \pi} - e^{-\sqrt{\lambda_k} \pi} = 0$$

$$e^{2\sqrt{\lambda_k} \pi} = 1 \Rightarrow 2\sqrt{\lambda_k} \pi = 2\pi i k$$

$$\sqrt{\lambda_k} = i k$$

$$\lambda_k = -k^2$$

Parametroje samo raznoje med α in β

$$w_k = e^{ikx} - e^{-ikx} \leadsto \text{BSS} \leadsto w_k = 2i \sin kx$$

$$\underline{w_k = \sin(kx)}$$

$$u(t, x) = \sum_{k=1}^{\infty} c_k e^{-k^2 t} \sin(kx)$$

$$u(0, x) = \sum_{k=1}^{\infty} c_k \sin(kx) = \sin x \xRightarrow{\text{keys}} c_k = 0 \text{ za } k \neq 1$$

$$u(t, x) = e^{-t} \sin x$$

MiDom de sau
manhele

①

$$u_{tt} - c^2 u_{xx} = F(t, x)$$

$$u_x(t, 0) = t^2$$

$$u(t, L) = -t$$

$$u(0, x) = x^2 - L^2$$

$$u_t(0, x) = \sin^2\left(\frac{\pi x}{L}\right)$$

u_1, u_2 rešitvi:

$$v = u_1 - u_2$$

$$v \equiv 0$$

$$\begin{aligned} v_{tt} - c^2 v_{xx} &= u_{1tt} - u_{2tt} - c^2 u_{1xx} + c^2 u_{2xx} = \\ &= F(t, x) - F(t, x) = 0 \end{aligned}$$

$$v_x(t, 0) = t^2 - t^2 = 0$$

$$v(t, L) = -t + t = 0$$

$$v(0, x) = 0$$

$$v_t(0, x) = 0$$

$$\varphi(t) = \int_0^L (v_t^2 + c^2 v_x^2) dx$$

$$\varphi'(t) = \int_0^L (2v_t v_{tt} + 2c^2 v_x v_{xt}) dx =$$

$$= 2 \int_0^L v_t v_{tt} dx + 2c^2 \int_0^L v_x v_{xt} dx$$

$$= 2 \int_0^L v_t v_{tt} dx + 2c^2 (v_x v_t \Big|_0^L - \int_0^L v_{xx} v_t dx) =$$

$$v_{tt} = c^2 v_{xx}$$

$$c^2 2 \int_0^L v_t v_{xx} + 2c^2 v_x v_t \Big|_0^L - 2c^2 \int_0^L v_{xx} v_t dx =$$

$$= 2c^2 (v_x(t, L) \cdot v_t(t, L) - v_x(t, 0) v_t(t, 0)) =$$

$$= 0$$

$$\begin{aligned} \Rightarrow \varphi(t) &= \varphi(0) = \int_0^L (v_t^2(0, x) + c^2 v_x^2(0, x)) dx = \\ &= \int_0^L (0 + c^2 \cdot 0) dx = 0 \end{aligned}$$

$$\Rightarrow v_t^2 + v_x^2 = 0 \Rightarrow v_x = 0 \quad v_t = 0$$

$$\Rightarrow v = C(y) \quad \wedge \quad v = C(x) \Rightarrow v = C$$

↑
konstanta

$$\Rightarrow v = 0 \text{ zaradi robnih pogojev}$$

What is a solution to PDE

$$F(x, u, \nabla u, \dots) = 0$$

A solution would be a function u s.t.

$$F(x, u, \nabla u, \dots) = 0$$

Since the equation involves derivatives of u up to some order $n \in \mathbb{N}$, we usually assume that the derivatives exist and are continuous

Def: A **classical solution** on an open set $\Omega \subseteq \mathbb{R}^2$ is $u \in C^n(\Omega)$ s.t. $F(x, u, \nabla u, \dots) = 0$

Standard questions:

- existence
- uniqueness
- maximal Ω

} usually related to the regularity of F

V 1. ora

7.5

$$u, v \in C^{1,2}((0, T) \times (0, L)) \cap C([0, T] \times [0, L])$$

$$u_t - u_{xx} \leq v_t - v_{xx} \quad \text{na } (0, T) \times (0, L)$$

$$u \leq v \quad \text{na parabolickém robu}$$

$$\{0\} \times [0, L] \cup [0, T] \times \{0, L\}$$

$$u \leq v \quad \text{na } [0, T] \times [0, L] = \Omega_T \quad \text{"}\Gamma_T$$

$$\text{Obecně vezmeme } u_t - v_t - u_{xx} + v_{xx} =$$

$$= (u - v)_t - (u - v)_{xx} \leq 0$$

$$\text{na robu: } u - v \leq 0$$

$$\Rightarrow \max_{\Omega_T \cup \Gamma_T} (u - v) = \max_{\Gamma_T \cup \partial} (u - v)$$

$$\max_{\Omega_T} (u - v) = \max_{\partial(\Omega_T)} (u - v) \leq 0$$

$$\Rightarrow u \leq v \quad \text{na } \Omega_T$$

Harmonične funkcije nima izoliranih ničel $n=2$

$$u_{xx} + u_{yy} = 0$$

$$u(x_0, y_0) = 0$$

$$u(x, y) = u(x + iy) = \frac{1}{2\pi} \int_{S_r} u(\alpha, \beta) dS_{\alpha, \beta}$$

če je izolirana ničla, obstaja dovolj mejna
delica, da nima ničel, torej u ne spreminja
predznaka

$$\frac{1}{2\pi r} \int_0^1 \underbrace{u(\gamma(t))}_{\neq 0} \underbrace{|\dot{\gamma}(t)|}_{> 0} dt \neq 0$$